

closure_evidence_20260925

BOB-179 — closure evidence

captured: 2026-09-25 branch: main

NOTE: no credential VALUE appears in this artifact or in any test fixture committed with it — variable NAMES only (§11.4.10). All test fixture values are synthetic placeholders (test-user-not-a-real-account, test-value-not-a-real-secret, SYNTHETIC-TEST-FIXTURE-VALUE), never real session material.

Scope

The BOB-172 independent review (2026-08-22/23) stated two gaps in the five-site refusal-not-reported-as-empty fix, both explicitly PRE-EXISTING and neither a regression from BOB-172:

- **Gap A** — an exception raised BEFORE `resp.status` is read is swallowed by each `_search_*` method's `except Exception: clause` (`logger.error; return results`) before `_search_one`'s own error handling can observe it. A connection-refused / DOWN tracker was reported identically to a genuinely empty one: `status="empty", error=None, http_status=None`.
- **Gap B** — a session-expiry redirect (302 -> login page -> 200) parses to zero rows and reports empty, because `_classify_upstream_http_status` correctly treats any 2xx as usable and all five private-tracker search GETs use aiohttp's default `allow_redirects=True`.

Both are now CLOSED as FIXES (not §11.4.112 structurally-impossible classifications — the item's own FIX DIRECTION text was correct that both were fixable, and no evidence emerged during investigation to the contrary).

Root cause / systematic-debugging (\$11.4.102)

Investigation confirmed both gaps genuinely present in download-proxy/src/merge_service/search.py as described, via direct source reading (not assumed from the prior review's own words):

- Gap A: `_search_rutracker` (both the `RUTRACKER_COOKIES` branch and the `RUTRACKER_USERNAME/RUTRACKER_PASSWORD` branch), `_search_nnmclub`, and `_search_iporrents` each had a bare `except Exception as e: logger.error(...)` with **no** diagnostic stash — confirmed by reading every one of the four methods' exception handlers before touching any of them. `_search_kinozal` was the ONE exception: its `except` clause already stashed a diagnostic, added earlier this same session as the BOB-235 fix for the identical defect class on kinozal specifically (confirmed by reading the file fresh per this item's own instruction not to assume prior state, since `kinozal/search.py` had been touched this session for BOB-137/BOB-235 before this item started).
- A SECOND call frame for the SAME defect class, not separately named by the review's Gap A text, was found during investigation: `_nnmclub_login`'s own `except Exception as e:` (the login-leg helper `_search_nnmclub` delegates to when `NNMCLUB_COOKIES` is unset and username/password auth is used) also swallowed the exception with no diagnostic — and a failure there never reaches `_search_nnmclub`'s own `except` at all, since `_nnmclub_login` never re-raises. Closed as part of this item's Gap A fix (see FIX below).
- Gap B: confirmed all five private-tracker search GETs (`_search_rutracker` cookie path, `_search_rutracker` credential path, `_search_kinozal`'s `browse.php` GET, `_search_nnmclub`'s `tracker.php` GET, `_search_iporrents`'s `/t` GET) call `session.get(...)` with no `allow_redirects` keyword — `aiohttp`'s default is `True`. Confirmed `_classify_upstream_http_status`'s own docstring and behaviour (`if 200 <= status < 300: return None`) is correct by design and was left untouched.

FIX

Gap A

Each of the four affected `except Exception:` clauses now stashes:

```

self._last_public_tracker_diag[<tracker>] = {
    "error_type": e.__class__.__name__,
    "error": f"<Tracker> request failed[: <leg>]: {e}",
    "stderr_tail": "",
    "deadline_hit": False,
    "deadline_seconds": 0.0,
}

```

— the EXACT dialect the BOB-235 kinozal fix (this session, prior to BOB-179) established for this defect class, reused verbatim rather than inventing a second one (§11.4.28). Applied at:

1. `_search_rutracker` — cookie-path except (RuTracker request failed (cookie path): ...)
2. `_search_rutracker` — credential-path except (RuTracker request failed: ...)
3. `_search_nnmclub` — search-leg except (NNMClub request failed: ...)
4. `_nnmclub_login` — login-leg except (NNMClub login request failed: ...) — the additional call frame found during investigation
5. `_search_ipstorrents` — except (IPTorrents request failed: ...)

`_search_kinozal` was already fixed (BOB-235) and is unchanged by this item.

Gap B

A new function, `_detect_session_expired_redirect(tracker_name, requested_url, history, final_url, final_status=None)`, added immediately after `_classify_upstream_http_status` in `search.py`:

- Returns None (no session-expiry signature) when history is empty/falsy — the overwhelming common case, and the one that matters most: it means NO redirect occurred, so a legitimate 200 search-results page that happens to mention the word “login” somewhere in its body never reaches any further check in this function at all.
- Returns None when a redirect DID occur but the final resolved path equals the requested path (e.g. an http→https upgrade landing back on the same search endpoint).
- Returns None when the final resolved path does not contain the substring “login” (the marker shared by every one of this module’s own login endpoints).
- Otherwise returns a diagnostic dict in the SAME shape `_classify_upstream_http_status` already returns (§11.4.28), with `error_type: "upstream_session_expired"`.

It is a **DISTINCT** detector — it inspects only `resp.history/resp.url` (authoritative aiohttp-level facts about what happened on the wire), never a body-text proxy, and it is **NOT** a widening of `_classify_upstream_http_status`'s status-code trigger: that function's own code, docstring, and test coverage are byte-for-byte unchanged by this item (diff-verified — see the RED/GREEN section below, where the negative-control tests pinning `_classify_upstream_http_status`'s existing behaviour stay green throughout).

Wired in at all five search-GET call sites, immediately AFTER `_check_search_response` (which correctly passes the response through — it's a 2xx) and BEFORE the row parser runs, capturing `resp.history` and `resp.url` while still inside the `async with block` (matching this file's own existing convention for capturing `resp.status`).

Stub reconciliation (§11.4.120)

Five sibling unit-test files' fake aiohttp-response stubs modelled an INCOMPLETE aiohttp contract — no `.history/.url` attributes — and started raising `AttributeError` once the Gap B wiring read them (an `AttributeError` inside the `try` block, itself then caught and diagnosed by the Gap A fix — which is how the incompleteness surfaced, rather than a silent skip). Per the SAME reconciliation discipline this log's earlier BOB-172 section already establishes (“the stub was wrong; the guard is right” — completing an aiohttp-contract field on a stub is not weakening a test, and making the PRODUCT code tolerate a missing attribute via `getattr(..., default)` was rejected there for a §11.4.252 fail-open reason that applies identically here):

- `tests/unit/merge_service/test_bob172_tracker_http_error_not_empty.py` — `_FakeResponse.__init__` gained `self.history = () / self.url = ""`.
- `tests/unit/merge_service/test_bob177_guard_wiring_collapse.py` — `_FakeResp.__init__`, same two lines.
- `tests/unit/merge_service/test_bob178_kinozal_login_status_diagnostic.py` — `_FakeResp.__init__`, same two lines.
- `tests/unit/merge_service/test_bob235_kinozal_connect_failure_diagnostic.py` — `_FakeResp.__init__`, same two lines.
- `tests/unit/merge_service/test_nnmclub_session_login.py` — `_FakeResp` already carried the BOB-172-era status-completion comment; extended with the identical `.history/.url` completion.

ZERO assertions were touched in any of the five files.

RED — observed BEFORE the fix (5 wiring call-sites + except-clause

diagnostics temporarily reverted; `_detect_session_expired_redirect` left defined but unwired, so it could not fire)

```
$ .venv/bin/python -m pytest tests/unit/merge_service/
test_bob179_stated_gaps_closure.py \
    -q --import-mode=importlib -p no:randomly
```

```
FAILED ...TestGapAConnectionRefusedSetsDiagnostic::test_rutracker_c
FAILED ...TestGapAConnectionRefusedSetsDiagnostic::test_rutracker_c
FAILED ...TestGapAConnectionRefusedSetsDiagnostic::test_nnmclub_sea
FAILED ...TestGapAConnectionRefusedSetsDiagnostic::test_nnmclub_log
FAILED ...TestGapAConnectionRefusedSetsDiagnostic::test_iporrents
FAILED ...TestGapBRedirectToLoginSetsDiagnostic::test_rutracker_coc
FAILED ...TestGapBRedirectToLoginSetsDiagnostic::test_rutracker_cre
FAILED ...TestGapBRedirectToLoginSetsDiagnostic::test_kinozal
FAILED ...TestGapBRedirectToLoginSetsDiagnostic::test_nnmclub
FAILED ...TestGapBRedirectToLoginSetsDiagnostic::test_iporrents
10 failed, 8 passed in 1.39s
```

Every failure is `AssertionError: ... diag is None` — exactly the BOB-172/BOB-172-review false-null signature the item names. The 8 that passed pre-fix are, by construction, the two negative-control classes (§11.4.201(1) — must be green both before AND after) plus the 4 isolated `_detect_session_expired_redirect` unit tests, which exercise the function directly and do not depend on any call-site wiring.

Method: the fix's 5 wiring call-sites and the 5 except-clause diagnostic stashes were reverted to their pre-BOB-179 text via the exact inverse of each edit applied (verified byte-identical to the intended pre-fix state by inspection of each reverted block); the file still compiled (`py_compile`) throughout. After capturing the RED run above, the fix was restored and a `diff` against a pre-revert backup copy of the file confirmed byte-identical restoration — no drift between the RED experiment and the shipped fix.

GREEN — after the fix (restored, diff-confirmed identical to the

intended shipped state)

```
$ .venv/bin/python -m pytest tests/unit/merge_service/
test_bob179_stated_gaps_closure.py \
    -q --import-mode=importlib -p no:randomly
```

18 passed in 1.28s

Negative controls (§11.4.201(1), mandatory — the over-fire direction)

Both gap classes carry their own, both green before AND after the fix:

- **Gap A:** `TestGapANegativeControlHealthyRoundTripStashesNothing` — a healthy round-trip through the rutracker credential path and through `_nnmclub_login`'s own username/password login leg (the two NEW call frames this item's fix touches) stashes NO diagnostic. The three other call frames (rutracker cookie path, kinozal, iptorrents) are already covered by the pre-existing sibling negative controls in `test_bob177_guard_wiring_collapse.py / test_bob235_kinozal_connect_failure_diagnostic.py`, whose healthy-path behaviour this item does not change.
- **Gap B:** `TestGapBNegativeControls` —
 1. `test_legitimate_200_mentioning_login_in_body_is_not_flagged`: a real 200 response with NO redirect (`resp.history` empty) whose body text merely mentions the word "login" must NOT be flagged — proves the detector keys on aiohttp's redirect bookkeeping, never a body scan;
 2. `test_redirect_that_lands_back_on_the_search_endpoint_is_not_flagged`: a redirect DID occur but the final path equals the requested search path — not a session-expiry signature, must NOT be flagged.

A guard that over-fired on either negative control would be a WORSE defect than the false-null being fixed (§11.4.201(1)) — it would report a healthy, working search as an error.

Full relevant suite (post-fix)

```
$ .venv/bin/python -m pytest tests/unit/merge_service/ \
    tests/unit/test_merge_trackers.py tests/unit/
test_private_tracker_search.py \
    -q --import-mode=importlib -p no:randomly
```

1159 passed in 160.66s (0:02:40)

ruff check on every file this item touched (search.py + the 6 test files):
all checks passed.

Files changed

- download-proxy/src/merge_service/search.py —
_detect_session_expired_redirect added; Gap A diagnostic-stashing added to _search_rutracker (both branches),
_search_nnmclub, _nnmclub_login, _search_iporrents; Gap B detector wired into all five private-tracker search-GET sites.
- tests/unit/merge_service/test_bob179_stated_gaps_closure.py —
new, 18 tests (this item's own RED/GREEN + negative controls + detector-unit coverage).
- tests/unit/merge_service/
test_bob172_tracker_http_error_not_empty.py,
test_bob177_guard_wiring_collapse.py,
test_bob178_kinozal_login_status_diagnostic.py,
test_bob235_kinozal_connect_failure_diagnostic.py,
test_nnmclub_session_login.py — stub reconciliation only
(.history = () / .url = "" added to each file's fake-response stub);
zero assertions touched.
- docs/qa/BOB-172/fix_evidence_20260822.log — stated-gaps closure
paragraph appended (both gaps marked CLOSED, summary of the
fix and proof, pointer to this file for full detail).
- docs/qa/BOB-179/closure_evidence_20260925.md — this file.

Not closed by this item (out of scope, unchanged)

- BOB-177, BOB-178, BOB-180 — cited in the 2026-08-23 stated-gaps
note as “also tracked from the same review, and likewise not closed
by this change” at the time; BOB-177 and BOB-178 have since been
closed separately (visible in docs/Fixed.md) — BOB-179 did not

touch or re-verify either. BOB-180 status was not investigated by this item.